UFCFK1-15-0 - Fundamentals of data science 25feb_1

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Documentation

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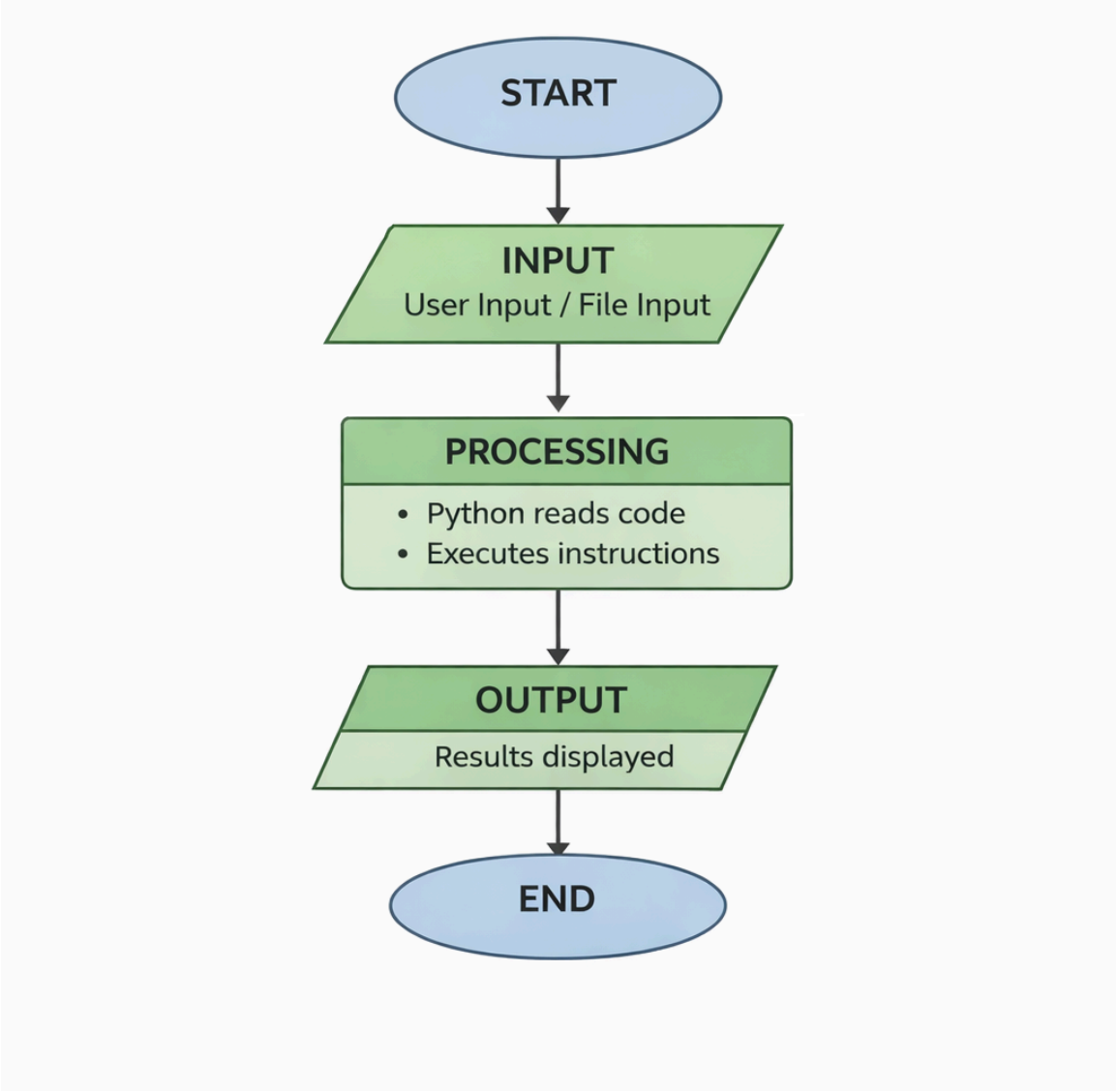
Section : A

GIT : https://github.com/999-ztj/FODS_assignments

Assignment 1

Question 1

The working of python



Question 2

Program Description:

```
This program performs basic arithmetic operations on two numbers.
```

Screenshot:

```
Enter first number: 6
Enter second number: 3
Addition: 9
Multiplication: 18
Division: 2.0
Modulus: 0
Exponentiation: 216
```

Question 3

Program Description:

```
This program checks whether a number is positive/negative and
even/odd.
```

Screenshot:

```
Enter a number: 2
Positive and Even
```

```
Enter a number: 3
Positive and Odd
```

```
Enter a number: -6
Negative and  Even
```

```
Enter a number: -17
Negative and Odd
```

```
Enter a number: 0
Zero
```

Question 4

Program Description:

```
This program performs cube, cube root, and logarithmic operations.
```

Screenshot:

```
Enter a number: 8
Cube: 512.0
Cube root: 2.0
Natural log: 2.0794415416798357
Log base 2: 3.0
Power 6: 262144.0
```

Question 5

Program Description:

```
This program calculates total, average, percentage and grade.
```

Screenshot:

```
Enter marks for 6 subjects (out of 100):
  Subject 1: 50
  Subject 2: 90
  Subject 3: 70
  Subject 4: 75
  Subject 5: 67
  Subject 6: 17

===== Result =====
Marks          : [50.0, 90.0, 70.0, 75.0, 67.0, 17.0]
Total          : 369.0 / 600
Average        : 61.50
Percentage     : 61.50%
Grade          : Second Division
```

Question 6

Program Description:

```
This program finds prime numbers in a given range.
```

Screenshot:

```
Enter start: 1
Enter end: 100
2
3
5
7
11
13
17
19
23
29
31
37
41
43
47
53
59
61
67
71
73
79
83
89
97
Count: 25
Sum: 1060
```

Question 7

Program Description:

```
This program finds numbers divisible by 9 but not by 6.
```

Screenshot:

```
Enter start: 1
Enter end: 100
9
27
45
63
81
99
```

1000 - 2500

Question 8

Program Description:

```
This program calculates the factorial of a positive integer.
```

Screenshot:

```
Enter a number: 5
Factorial: 120
```

```
Enter a number: -2
Invalid input! Please enter a positive integer.
```

Question 9

Program Description:

```
This program sums positive and negative numbers separately.
```

Screenshot:

```
Enter number: 17
Continue? (y/n): y
Enter number: 19
Continue? (y/n): y
Enter number: -17
Continue? (y/n): y
Enter number: -19
Continue? (y/n): n
Positive sum: 36
Negative sum: -36
```

Question 10

Program Description:

```
This program is a number guessing game.
```

Screenshot:

```
Enter your guess: 50
Too high
Enter your guess: 25
Too high
Enter your guess: 15
Too high
Enter your guess: 10
Too high
Enter your guess: 5
Too high
Enter your guess: 3
Too high
Enter your guess: 2
Correct!
```

Assignment 2

Question 1

Program Description:

This program uses a function to calculate sum, difference, product, and quotient.

Screenshot:

```
Enter first number: 1
Enter second number: 2
Sum: 3
Difference: -1
Product: 2
Quotient: 0.5
```

Question 2

Program Description:

This program performs mathematical operations using separate functions.

Screenshot:

```
Enter first number: 1
Enter second number: 2
Addition: 3
Multiplication: 2
Division: 0.5
Floor Division: 0
Exponentiation: 1

Process finished with exit code 0
```

Question 3

Program Description:

```
This program checks whether a number is an Armstrong number.
```

Screenshot:

```
Enter a number: 153
Armstrong number
```

Question 4

Program Description:

```
This program counts the frequency of numbers in a list.
```

Screenshot:

```
{1: 1, 2: 2, 3: 1}
{4: 3, 5: 1}
{7: 2, 8: 1, 9: 1}
```

Question 5

Program Description:

```
This program sorts student names in reverse alphabetical order.
```

Screenshot:

```
Enter names separated by comma: Jenith,Sara
Sorted names: Sara Jenith
```

Question 6

Program Description:

This program searches for a country in a list.

Screenshot:

```
Enter countries separated by comma: Nepal,Christmas_islands,Georgia,Mongolia
Enter country to search: Christmas_islands
1
```

Question 7

Program Description:

This program counts characters in a sentence excluding spaces.

Screenshot:

```
Enter a sentence: God is good God is great
{'G': 2, 'o': 4, 'd': 3, 'i': 2, 's': 2, 'g': 2, 'r': 1, 'e': 1, 'a': 1, 't': 1}
```

Question 8

Program Description:

This program removes duplicates and sorts numbers.

Screenshot:

```
Enter numbers separated by space: 2 2 2 2 2 2 2 2 2 5 6 7 1 2 1 3 4 566 66 4 4 3
Result: [1, 2, 3, 4, 5, 6, 7, 66, 566]

Process finished with exit code 0
```

Question 9

Program Description:

```
This program stores details of 3 movies in a dictionary.
```

Screenshot:

```
Enter movie title: Marry my Husband
Enter director: Saranshika Kc
Enter release year: 2026
Enter rating: 9
Enter movie title: Friren
Enter director: Jenith Gurung
Enter release year: 2026
Enter rating: 10
Enter movie title: Maze runner
Enter director: Sara kc
Enter release year: 2024
Enter rating: 3

Title: Marry my Husband
Director: Saranshika Kc
Year: 2026
Rating: 9

Title: Friren
Director: Jenith Gurung
Year: 2026
Rating: 10

Title: Maze runner
Director: Sara kc
Year: 2024
Rating: 3

Process finished with exit code 0
```

Question 10

Program Description:

```
This program is a dice guessing game.
```

Screenshot:

```
Guess the dice value (1-6): 5
😐 Close!Dice was: 4
```

```
Guess the dice value (1-6): 1
❌ Wrong! Dice was: 5
```

```
Guess the dice value (1-6): 6
😊 Correct!
```

Assignment 3 - File Handling & OOP

Question 1

Program Description:

This program reads the file "records.csv" and displays its contents.The fields are: student_name, roll_no, program, year, and group.

Screenshot:

```
=== Student Records ===

student_name,roll_no,program,year,group
--------------------------------------------------
Name         : Ivan Tamrakar
Roll No      : TBC001
Program      : BSc AI
Year         : 2
Group        : A
--------------------------------------------------
Name         : Aayatish Gautam
Roll No      : TBC002
Program      : BSc CS
Year         : 1
Group        : B
--------------------------------------------------
Name         : Jasmine karki
Roll No      : TBC003
Program      : BIT
Year         : 3
Group        : A
--------------------------------------------------
Name         : Jenith Gurung
Roll No      : TBC004
Program      : BSc AI
Year         : 2
Group        : C
--------------------------------------------------
Name         : Saranshika Kc
Roll No      : TBC005
Program      : BCA
Year         : 1
Group        : A
--------------------------------------------------
```

Question 2

Program Description:

This program takes student details as input from the user and appends them to the existing file "records.csv".Fields: student_name, roll_no, program, year, and group.

Screenshot:

```
=== Add New Student Record ===

Enter student name: Jenith.jr Gurung
Enter roll number: TBC017
Enter program: Bsc AI
Enter year: 1
Enter group: A

Record added successfully!
Added: Jenith.jr Gurung,TBC017,Bsc AI,1,A
```

	student_name	roll_no	program	year	group
1	Ivan Tamrakar	TBC001	BSc AI	2	A
2	Aayatish Gautam	TBC002	BSc CS	1	B
3	Jasmine karki	TBC003	BIT	3	A
4	Jenith Gurung	TBC004	BSc AI	2	C
5	Saranshika Kc	TBC005	BCA	1	A
6	Jenith.jr Gurung	TBC017	Bsc AI	1	A

Question 3

Program Description:

This program accepts two integers from the user, performs addition, subtraction, multiplication, and division, and saves results to a file with the current date and time. It keeps running until the user exits.

Screenshot:

```
=== Calculator with File Saving ===
Type 'exit' when asked for first number to quit.

Enter first number (or 'exit' to quit): 17
Enter second number: 19
Results saved at 2026-04-17 19:44:13

Enter first number (or 'exit' to quit): exit

=== Saved Results ===
Date/Time  : 2026-04-17 19:44:13
Numbers    : 17 and 19
Addition   : 36
Subtraction: -2
Multiply   : 323
Division   : 0.8947368421052632
----------------------------------------
```

q3_operations_file.py results.csv

Edit in Data Wrangler

	C1
1	Date/Time : 2026-04-17 21:07:40
2	Numbers : 17 and 19
3	Addition : 36
4	Subtraction: -2
5	Multiply : 323
6	Division : 0.8947368421052632
7	----------

Question 4

Program Description:

This program reads content from one file and copies it into another file.The user enters both file names. It uses try/except to handle errors such as an input file not found or an output file already existing.

Screenshot:

```
=== File Copy Program ===

Enter the name of the input file: data.txt
Enter the name of the output file: copy.txt

Error: Input file does not exist.
[Errno 2] No such file or directory: 'data.txt'

Process finished with exit code 0
```

```
=== File Copy Program ===

Enter the name of the input file: test.txt
Enter the name of the output file: copy.txt

File copied successfully from 'test.txt' to 'copy.txt'.

Process finished with exit code 0
```

Question 5

Program Description:

This program creates a class called Learner with attributes: roll_no, full_name, address, enrollment_year, program, and group.It takes input for all attributes and displays the details.

Screenshot:

```
Enter Learner Details:

Roll No          : 17
Full Name        : Jenith Gurung
Address          : Koteshwor
Enrollment Year  : 2025
Program          : Bsc AI
Group            : A

=== Learner Details ===
Roll No          : 17
Full Name        : Jenith Gurung
Address          : Koteshwor
Enrollment Year  : 2025
Program          : Bsc AI
Group            : A

Process finished with exit code 0
```

Question 6

Program Description:

This program creates a Dish class with attributes: dish_id, dish_name,ingredients, and instructions. A Cookbook class manages a collection of Dish objects with options to add and display dishes.

Screenshot:

```
=== Add Dishes to Cookbook ===

Do you want to add a dish? (yes/no): yes
  Dish ID       : 1719
  Dish Name     : Bibimbap
  Ingredients   : Rice,gochujang,veggies
  Instructions  : mix all ingredients together
Dish 'Bibimbap' added to cookbook.

Do you want to add a dish? (yes/no): yes
  Dish ID       : 1917
  Dish Name     : pasta
  Ingredients   : pasta,veggies
  Instructions  : boil it
Dish 'pasta' added to cookbook.

Do you want to add a dish? (yes/no): no

=== All Dishes in Cookbook ===
  Dish ID       : 1719
  Dish Name     : Bibimbap
  Ingredients   : Rice,gochujang,veggies
  Instructions  : mix all ingredients together

  Dish ID       : 1917
  Dish Name     : pasta
  Ingredients   : pasta,veggies
  Instructions  : boil it
```

Question 7

Program Description:

This program creates a Staff class with attributes: emp_id, full_name, address, phone_number, marital_status, dependents, and salary.It allows adding multiple staff members, saves them to staff.csv, and displays the list of staff. Uses try/except for error handling.

Screenshot:

```
=== Staff Management System ===

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 1

Employee ID    : 1719
Full Name      : Jenith Gurung
Address        : KTM
Phone Number   : 9765828069
Marital Status : Married
Dependents     : 2
Salary         : 100000
Staff data saved to staff.csv

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 2

=== Staff List ===
------------------------------------------------------------
emp_id, full_name, address, phone_number, marital_status, dependents, salary
1719, Jenith Gurung, KTM, 9765828069, Married, 2, 100000
------------------------------------------------------------

1. Add Staff Member
2. View All Staff
3. Exit
Enter choice: 3
Exiting program.

Process finished with exit code 0
```

Question 8

Program Description:

This program is a basic Library Management System.It allows users to: borrow a book, return a book, and search for a book.It uses OOP (classes), file handling for storing book data, and try/except for exception handling.

Screenshot:

```
=== Library Management System ===

1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 4

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Available
------------------------------------------------------------
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 1
Enter book title or author to search: Machine Learning

Search results for 'Machine Learning':
--------------------------------------------------
ID: B003 | Title: Machine Learning | Author: Jenith Gurung | Available
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 2

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Available
------------------------------------------------------------
Enter Book ID to borrow: B004
You have borrowed 'Database Systems'. Please return it on time.
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 3

=== All Books ===
------------------------------------------------------------
ID: B001 | Introduction to Python | John Smith | Available
ID: B002 | Data Science Basics | Surav joshi | Available
ID: B003 | Machine Learning | Jenith Gurung | Available
ID: B004 | Database Systems | Sara kc | Borrowed
------------------------------------------------------------
Enter Book ID to return: B004
Thank you for returning 'Database Systems'.
```

```
1. Search for a Book
2. Borrow a Book
3. Return a Book
4. View All Books
5. Exit

Enter your choice: 5
Thank you for using the Library System. Goodbye!

Process finished with exit code 0
```

Assignment 4

Question 1

Program Description:

This program generates a NumPy array and performs basic operations: finding the sum, mean, largest and smallest values.

Screenshot:

```
Array: [10 20 30 40 50]

Sum of all elements: 150
Mean value: 30.0
Largest value: 50
Smallest value: 10
```

Question 2

Program Description:

This program takes a list of at least 12 numbers from the user,sorts the list, and performs slicing to extract elements at index ranges 3-6, 6-9, and 4-10.

Screenshot:

```
Enter at least 12 numbers separated by spaces:
1 2 3 4 5 6 7 8 9 10 11 12

Original list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]
Sorted list: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]

Slice [3:6] (index 3 to 5): [4, 5, 6]
Slice [6:9] (index 6 to 8): [7, 8, 9]
Slice [4:10] (index 4 to 9): [5, 6, 7, 8, 9, 10]

Process finished with exit code 0
```

Question 3

Program Description:

```
This program creates an array of 12 random integers using
NumPy,sorts the array, and reshapes it into a 3x4 matrix.
```

Screenshot:

```
Original random array: [46  5 45 37 38 81 65 74 13 56 24 56]
Sorted array: [ 5 13 24 37 38 45 46 56 56 65 74 81]

Reshaped into 3x4 matrix:
[[ 5 13 24 37]
 [38 45 46 56]
 [56 65 74 81]]

Reshaped into 4x3 matrix:
[[ 5 13 24]
 [37 38 45]
 [46 56 56]
 [65 74 81]]
```

Question 4

Program Description:

Matrix operations using NumPy with dimension validation.Performs addition, subtraction, and multiplication. Raises exceptions if dimensions are incompatible.

Screenshot:

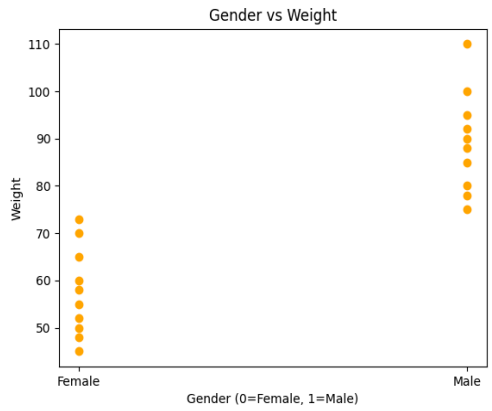
```
=== Matrix Operations ===

Enter rows for Matrix 1: 2
Enter columns for Matrix 1: 2
Enter 4 elements for Matrix 1:
1 2 3 4

Enter rows for Matrix 2: 2
Enter columns for Matrix 2: 2
Enter 4 elements for Matrix 2:
5 6 7 8

Matrix 1:
 [[1 2]
 [3 4]]

Matrix 2:
 [[5 6]
 [7 8]]

Addition Result:
 [[ 6  8]
 [10 12]]

Subtraction Result:
 [[-4 -4]
 [-4 -4]]

Multiplication Result:
 [[19 22]
 [43 50]]
```

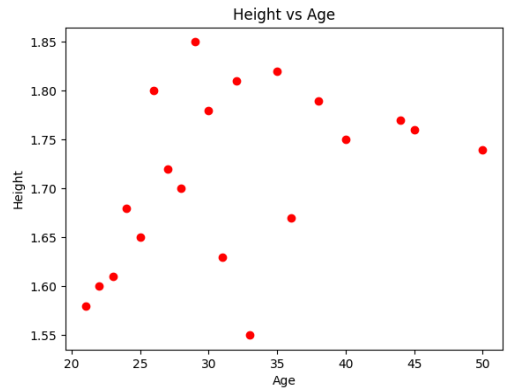
Question 5

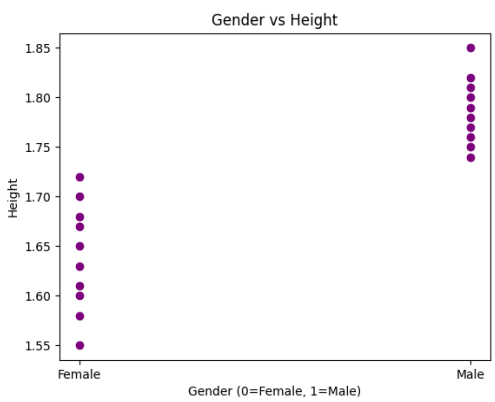
Program Description:

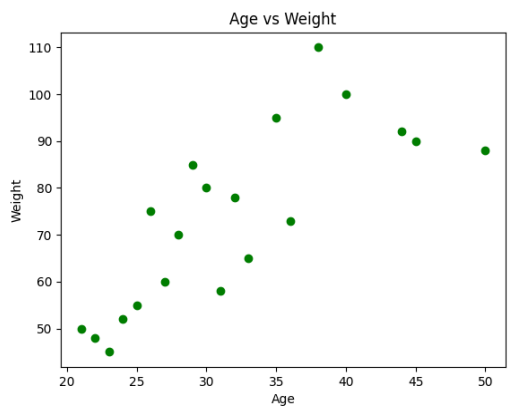
This program reads the file health_data.csv using Pandas.It creates scatter plots to show relationships between variables such as height, weight, age, and gender using Matplotlib.The plots are displayed and saved as image files.

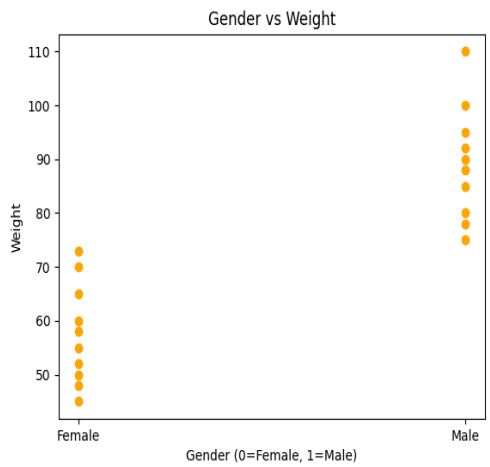
Screenshot:

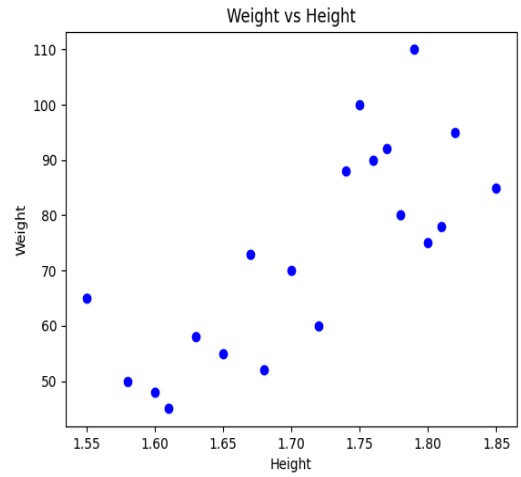













```
Data loaded successfully.
First 5 rows:
     name  age  gender  height  weight
0   Alice   25  Female    1.65      55
1  Jenith   30    Male    1.78      80
2   Carol   22  Female    1.60      48
3   David   35    Male    1.82      95
4    Emma   28  Female    1.70      70

Plot 1 saved: weight vs height
Plot 2 saved: age vs weight
Plot 3 saved: height vs age
Plot 4 saved: gender vs height
Plot 5 saved: gender vs weight

All plots created successfully.

Process finished with exit code 0
```

Question 6

Program Description:

This program reads health_data.csv into a Pandas DataFrame and adds two new columns: BMI (Weight / Height) and Health_Status based on the BMI value ranges provided.

Screenshot:

```
Original data (first 5 rows):
     name  age  gender  height  weight
0   Alice   25  Female    1.65      55
1  Jenith   30    Male    1.78      80
2   Carol   22  Female    1.60      48
3   David   35    Male    1.82      95
4    Emma   28  Female    1.70      70

Updated data with BMI and Health_Status (first 10 rows):
     name  age  gender  ...  weight    BMI                        Health_Status
0   Alice   25  Female  ...      55  33.33  High risk of diabetes/heart disease
1  Jenith   30    Male  ...      80  44.94            Critical health condition
2   Carol   22  Female  ...      48  30.00  High risk of diabetes/heart disease
3   David   35    Male  ...      95  52.20            Critical health condition
4    Emma   28  Female  ...      70  41.18            Critical health condition
5   Frank   40    Male  ...     100  57.14            Critical health condition
6   Grace   31  Female  ...      58  35.58            Critical health condition
7   Henry   26    Male  ...      75  41.67            Critical health condition
8    Isla   24  Female  ...      52  30.95  High risk of diabetes/heart disease
9    Jack   45    Male  ...      90  51.14            Critical health condition

[10 rows x 7 columns]

Updated data saved to health_data_updated.csv

Health Status Summary:
Health_Status
Critical health condition              14
High risk of diabetes/heart disease     5
Overweight risk                         1
Name: count, dtype: int64

Process finished with exit code 0
```